CIPHERPASS - BANKING KYC & AML CUSTOMER SCREENING SYSTEM

DIGITAL IMAGE PROCESSING

Presented By:

- Jumaira Osman (15)
- Maria Atta (01)
- Azka Nasir (26)

Presented To:

- Ma'am Sana

Introduction

CipherPass is a banking verification system that uses Digital Image Processing (DIP) and Computer Vision to automate KYC and AML screening. The system verifies CNIC authenticity, performs face verification and liveness detection, checks criminal records, and generates a trust score. It helps banks reduce identity fraud, improve security, and make faster verification decisions.

Problem

- Manual KYC verification is slow and time-consuming.
- Fake CNICs and forged documents lead to identity fraud.
- Criminal and AML screening is often not integrated into one system.
- Banks face difficulties in detecting duplicate or suspicious verifications.

Solution

- Develop an automated KYC and AML verification system.
- Use Digital Image Processing to analyze and enhance CNIC images.
- Perform face verification and liveness detection for identity confirmation.
- Generate a trust score and risk level to support secure banking decisions.

System Architecture & Workflow

User Input Module

- CNIC Upload
- Face Image Upload
- Customer Information Entry

Database & AML Screening Module

- Citizen Database Lookup
- Criminal Record Check
- Duplicate Verification Detection

DIP Processing Module

- Grayscale Conversion
- Histogram Equalization
- CLAHE Enhancement
- Edge Detection

System Architecture & Workflow

Risk Engine & Dashboard

- Trust Score Calculation
- Risk Classification
- Approve / Review / Reject Decision

Face Verification Module

- Haar Cascade Face Detection
- Liveness Detection using Laplacian Variance

Core Concepts

CONCEPT	PURPOSE	USE
Grayscale Conversion	Simplify image processing	CNIC and face image preprocessing
Histogram Equalization	Improve image contrast	Enhance low-light CNIC images
CLAHE	Local contrast enhancement	Improve text visibility on CNIC
Gaussian Blur	Reduce image noise	Preprocessing before edge detection
Canny Edge Detection	Detect boundaries and edges	CNIC border and forgery detection
Perspective Transform	Correct image distortion	Deskew tilted CNIC images
Haar Cascade	Detect human faces	Face verification module
Laplacian Variance	Measure image sharpness	Liveness detection and blur analysis

Conclusion

- Successfully developed an automated KYC and AML verification system for banking applications.
- Implemented Digital Image Processing techniques for CNIC enhancement and analysis.
- Integrated face verification, liveness detection, and criminal record screening.
- Generated trust-based risk assessment to support secure decision-making.
- Reduced the risk of identity fraud while improving verification efficiency and compliance.

THANK YOU